

Scheduler algorithm comparison

Retained field validation, 29 September 2026 | Technical decision brief

Decision context. Modelled cost is the lead measure, subject to served demand, feasibility, runtime, and evidence quality. Keep the current booking selection and daily TABU default while evaluating candidates. These fixture results do not establish production savings.

Evidence scope

The retained matrix has **218 validated booking cases** (12 original, 30 screen, 144 held out, 8 stress, 24 browser) and **240 daily solver cases**. The two held-out seeds are 59 and 83. Original booking cases ran under an earlier bundled policy; subsequent cases use the four-hour, one-offer, zero-new-overtime policy. Daily roads are deterministic directed fixture legs; held-out dispersed booking roads use GraphHopper 11.

Booking: cost with service and runtime

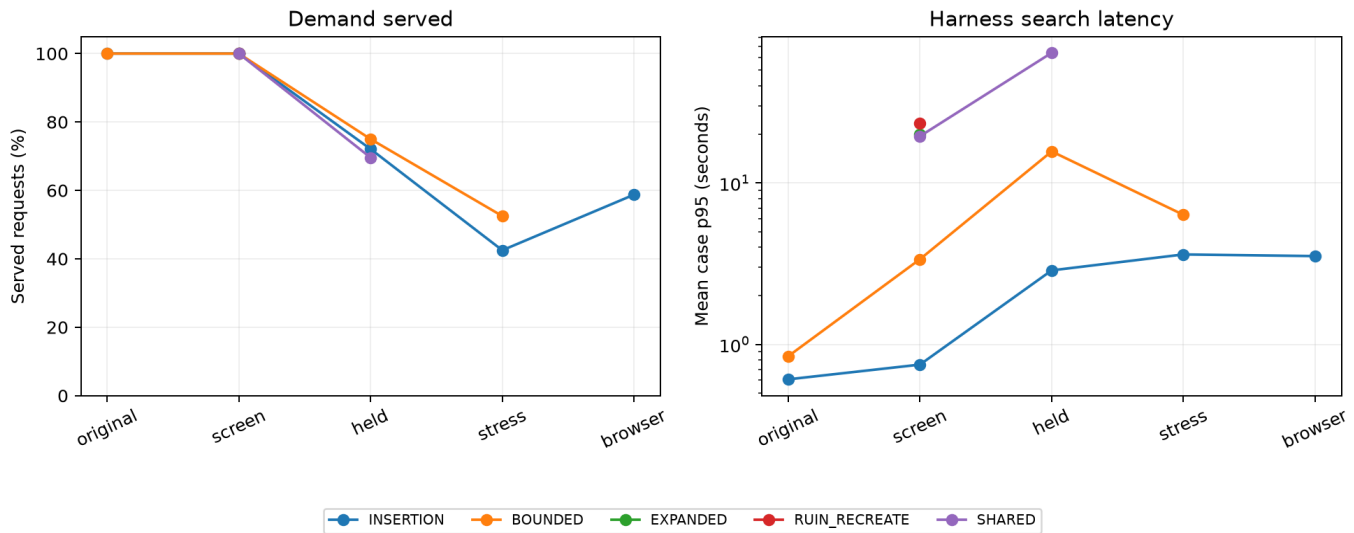


Figure 1. Served rate and latency are shown only where each variant was tested. Lines connect separate experiment stages for readability and do not imply a time trend. The original policy is not an isolated algorithm comparison. The browser point is insertion only; stress has insertion and bounded only. Held-out SHARED served 334/480 against insertion 346/480 and bounded 360/480, with 64.1 s mean case p95 and 153 incomplete searches.

The two panels answer different operational questions. The served panel counts confirmed requests; the latency panel averages each case's p95 search latency and uses a log scale because the slowest configuration is more than twenty times the fastest held-out option. A lower modeled route cost has little value if the search times out or leaves demand unserved. The held-out, stress, and browser points should be read as distinct workloads with different coverage, not as a single progression.

Held-out demand and completion

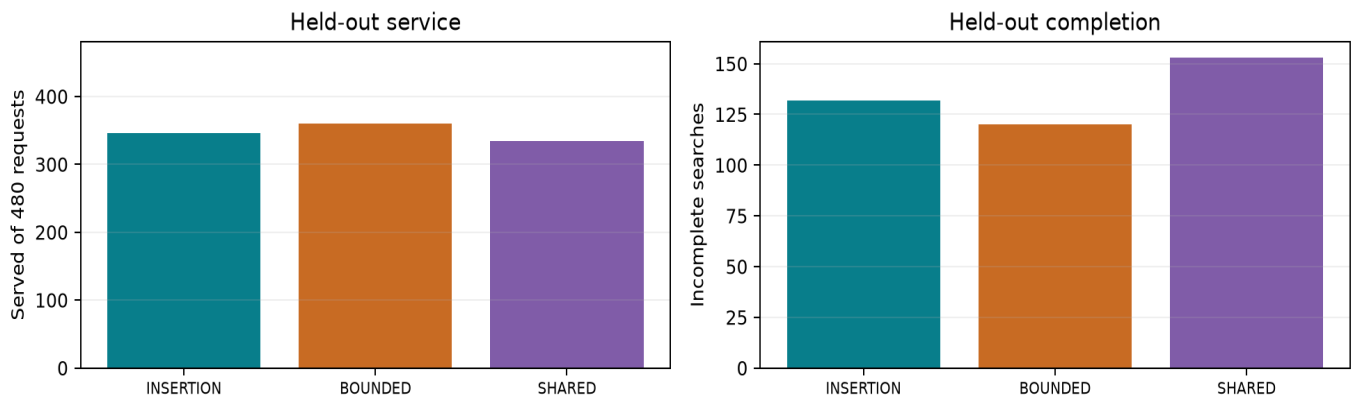


Figure 1b. Each held-out variant has 48 cases and 480 attempted requests. Bars separate successful service from incomplete searches, which are not equivalent to proven infeasibility. BOUNDED serves 14 more requests than INSERTION, while SHARED serves 12 fewer. SHARED also leaves 33 more searches incomplete than BOUNDED, so its matched-cost result needs service and completion context.

Booking cost: matched customers only

Stage	Variant vs insertion	Matched / changed cases	Served delta	Mean savings per stream	Exploratory 95% interval
screen	BOUNDED	6/0	0	\$12.50	n/a
screen	EXPANDED	6/0	0	\$15.00	n/a
screen	RUIN_RECREATE	6/0	0	\$20.00	n/a
screen	SHARED	6/0	0	\$20.00	n/a
held	BOUNDED	15/33	14	\$23.44	\$23.28 to \$23.59
held	EXPANDED	0/0	0	n/a	n/a
held	RUIN_RECREATE	0/0	0	n/a	n/a
held	SHARED	16/32	-12	\$31.36	\$29.84 to \$32.87

Final inventory cost includes existing work and different served customers, so its totals cannot be read as savings. The table uses only cases with identical served request indices and reports changed-customer cases separately. Screen cases collapse to one clustered request stream; held-out intervals use only two independent seeds and are exploratory.

Comparable cost by customer matched stream

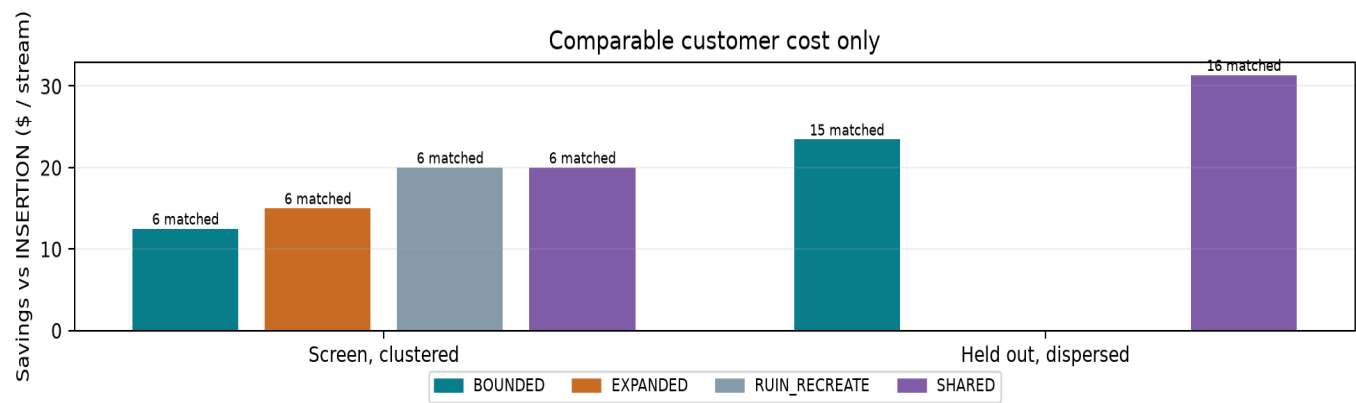


Figure 2. Bars show savings per matched stream for each tested variant. Left to right within each stage: BOUNDED, EXPANDED, RUIN_RECREATE, SHARED. The absent held-out EXPANDED and RUIN_RECREATE bars mean untested, not zero. The numbers above bars are matched case counts; differing served identities are excluded.

The screen bars are a controlled cost signal with all 18 requests served per variant. Their one clustered stream has repeated geography, so the differences may reflect cleanup of modeled waiting and buffer time rather than a repeatable field saving. The held-out bars use dispersed road travel and show larger savings among matched cases, but BOUNDED and SHARED changed customer identity in 33 and 32 of 48 cases. Figure 2 therefore cannot rank total customer outcomes by cost alone.

Search quality over time

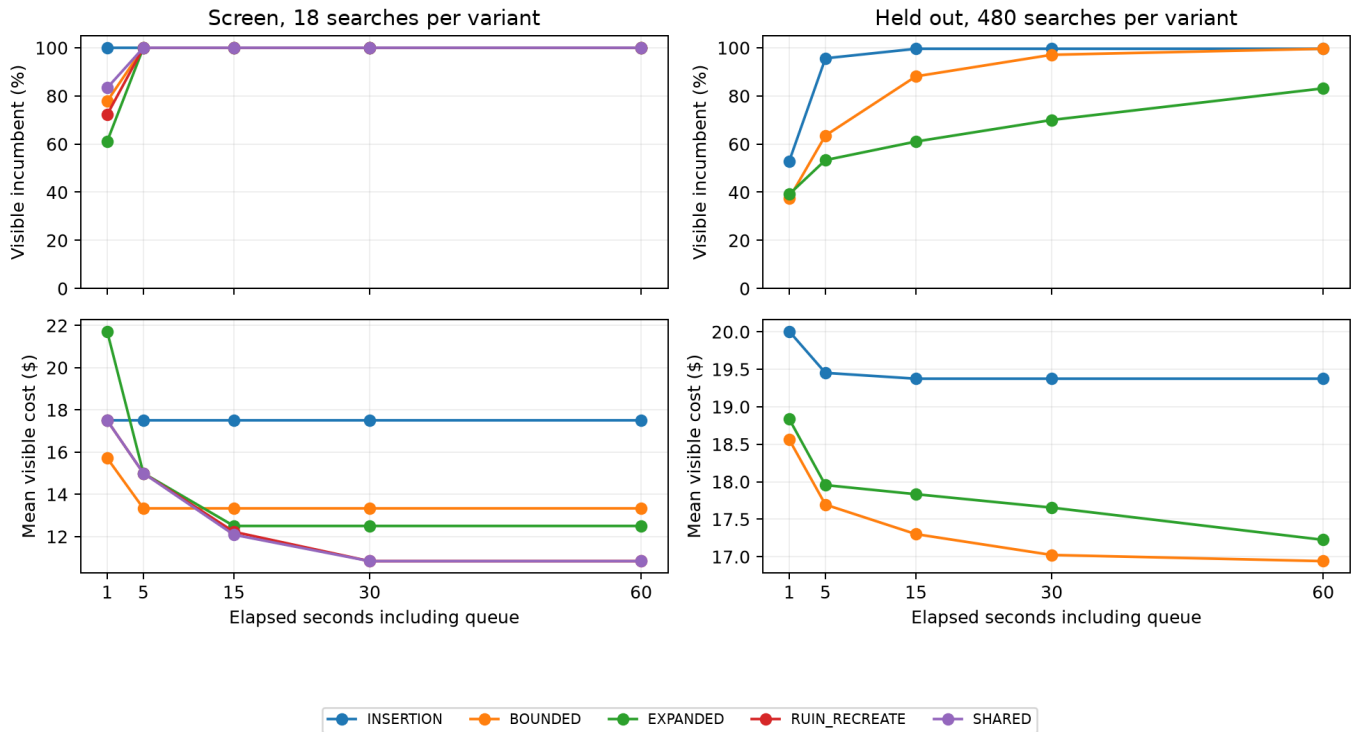


Figure 3. Upper panels show visibility by checkpoint; lower panels show mean incremental cost among visible incumbents. Missing incumbents stay missing and are excluded from the lower means. A visible incumbent can still be invalidated at final revalidation. At 60 s, held-out visibility was 478/480 for insertion and bounded, and 399/480 for SHARED.

The upper held-out panel shows INSERTION finding a visible option sooner than BOUNDED, while BOUNDED's visible cost falls further by later checkpoints. SHARED finds fewer visible options throughout the minute. The lower panels describe the quality of options already visible, so their means have changing denominators; a downward line does not prove an all-request cost improvement. Queue time and roughly one-second polling are included, making this a customer-visible progress measure rather than exact solver time-to-best.

Daily solver: accepted cost by budget

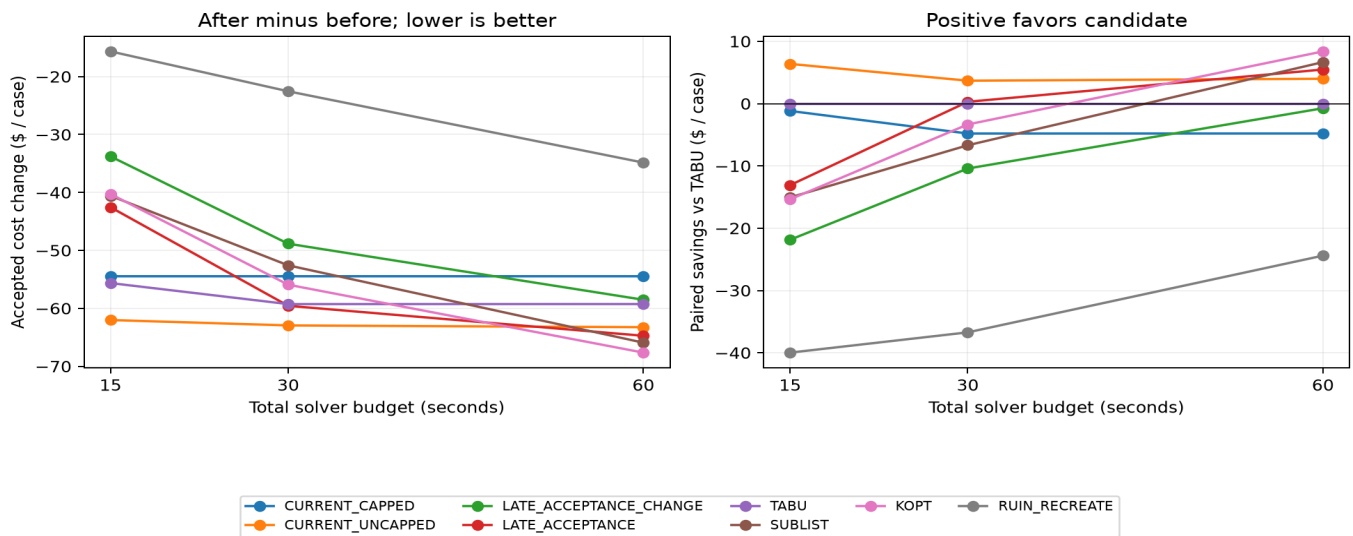


Figure 4. Left: mean accepted cost change from the before snapshot. Right: paired savings against TABU on the same fingerprint and budget, averaging fleet sizes inside each seed. All 240 daily results have zero reported violations. Capped runs end in about 3 seconds; other held-out runs consume approximately their 15, 30, or 60 second budget.

The left plot answers whether a configuration improved its own starting schedule; every line remains below zero. The right plot answers which configuration produced a lower final cost on the same held-out dataset and time budget. Those rankings differ because starting costs and case mix matter. CURRENT_UNCAPPED is favorable at each tested budget, while KOPT, SUBLIST, and LATE_ACCEPTANCE cross above TABU only at 60 seconds. RUIN_RECREATE improves its own schedule as time rises but remains behind TABU on paired final cost.

Held-out daily measurements: 15 and 30 seconds

Configuration	Budget s	Mean accepted cost change	Mean elapsed s	Paired savings vs TABU \$
CURRENT_CAPPED	15	\$-54.46	3.2	-1.18
CURRENT_UNCAPPED	15	\$-62.02	15.0	+6.38
LATE_ACCEPTANCE_CHANGE	15	\$-33.78	15.0	-21.85
LATE_ACCEPTANCE	15	\$-42.55	15.0	-13.09
TABU	15	\$-55.64	15.0	0
SUBLIST	15	\$-40.58	15.0	-15.06
KOPT	15	\$-40.28	15.0	-15.36
RUIN_RECREATE	15	\$-15.66	15.0	-39.98
CURRENT_CAPPED	30	\$-54.46	2.9	-4.80
CURRENT_UNCAPPED	30	\$-62.94	30.0	+3.68
LATE_ACCEPTANCE_CHANGE	30	\$-48.84	30.0	-10.42
LATE_ACCEPTANCE	30	\$-59.56	30.0	+0.30
TABU	30	\$-59.26	30.0	0
SUBLIST	30	\$-52.58	30.0	-6.68
KOPT	30	\$-55.92	30.0	-3.35
RUIN_RECREATE	30	\$-22.54	30.0	-36.72

Held-out daily measurements: 60 seconds

Configuration	Budget s	Mean accepted cost change	Mean elapsed s	Paired savings vs TABU \$
CURRENT_CAPPED	60	\$-54.46	2.8	-4.79
CURRENT_UNCAPPED	60	\$-63.25	60.0	+3.99
LATE_ACCEPTANCE_CHANGE	60	\$-58.52	60.0	-0.74
LATE_ACCEPTANCE	60	\$-64.73	60.0	+5.48
TABU	60	\$-59.26	60.0	0
SUBLIST	60	\$-65.92	60.0	+6.67
KOPT	60	\$-67.64	60.0	+8.38
RUIN_RECREATE	60	\$-34.85	60.0	-24.41

Cost change is after minus before, so more negative means more modeled cleanup. The paired column is the sounder comparison with TABU. Each configuration has eight held-out cases per budget: two seeds times four fleet sizes. The 15-second screen adds six cases per configuration, yielding 30 cases each and 240 total.

Exploratory paired intervals

Configuration	15 s	30 s	60 s
CURRENT_CAPPED	-1.18 [-5.15, +2.80]	-4.80 [-8.00, -1.60]	-4.79 [-8.00, -1.59]
CURRENT_UNCAPPED	+6.38 [+2.35, +10.41]	+3.68 [-0.51, +7.88]	+3.99 [+0.05, +7.93]
LATE_ACCEPTANCE_CHANGE	-21.85 [-25.66, -18.05]	-10.42 [-10.60, -10.24]	-0.74 [-2.31, +0.83]
LATE_ACCEPTANCE	-13.09 [-16.25, -9.92]	+0.30 [-1.11, +1.71]	+5.48 [+3.26, +7.70]
TABU	baseline	baseline	baseline
SUBLIST	-15.06 [-16.82, -13.30]	-6.68 [-8.30, -5.07]	+6.67 [+5.13, +8.20]
KOPT	-15.36 [-17.73, -12.98]	-3.35 [-5.75, -0.94]	+8.38 [+5.59, +11.17]
RUIN_RECREATE	-39.98 [-42.82, -37.14]	-36.72 [-39.43, -34.02]	-24.41 [-25.02, -23.80]

Values are dollars per case. The percentile bootstrap resamples only two independent held-out seed units after averaging fleet sizes; its intervals are exploratory and should not be interpreted as population confidence. More complex neighborhoods at different acceptor settings do not isolate the causal effect of a single move.

Runtime and feasibility by configuration

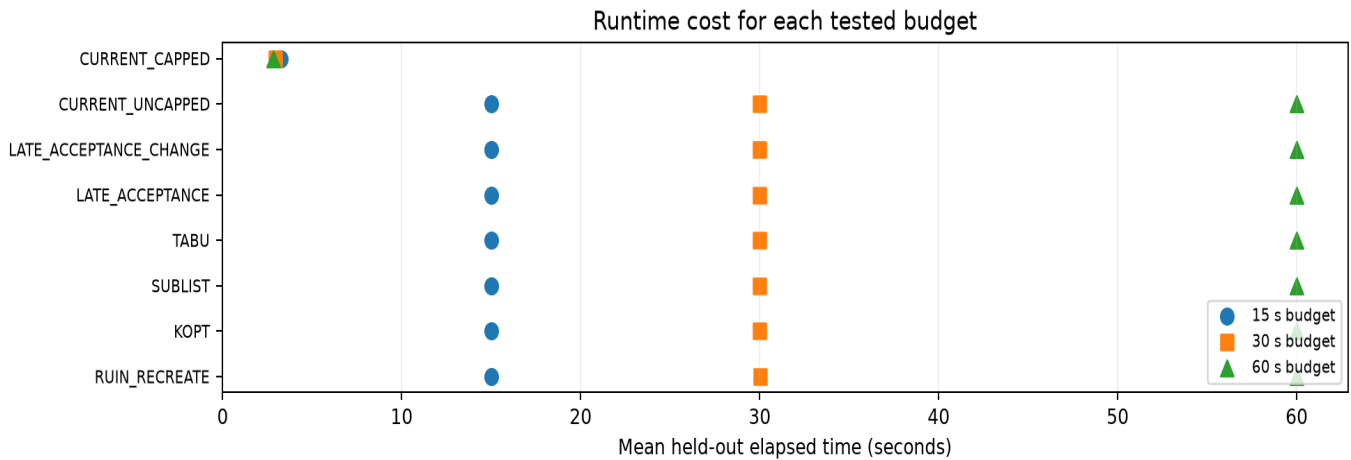


Figure 5. Each marker is the mean elapsed time of eight held-out cases for one configuration and budget. `CURRENT_CAPPED` stops near three seconds regardless of allowance; other configurations run for the allocated 15, 30, or 60 seconds. The validation reports zero violations in all 240 daily cases. This is a fixture feasibility result, not proof for every future schedule.

Booking algorithm comparison

INSERTION

This is the policy-only search control: it tries eligible routes and insertion positions without rearranging existing visits. It is the quickest held-out option at 2.9 seconds mean case p95 and served 346 of 480 requests. Its 19.37 dollar visible incremental cost at 60 seconds is above `BOUNDED`'s 16.94 dollars among visible searches. Those visible-cost means condition on visibility, while the matched-case table provides the fairer final cost comparison.

BOUNDED

Relocation, swap, and reversal are limited to six routes, depth two, beam eight, and 500 arrangements per window. It served 360 of 480 held-out requests, 14 more than `INSERTION`, and yielded 23.44 dollars mean savings per matched stream. That result comes with 15.7 seconds mean case p95 and 120 incomplete searches. Only 15 of 48 held-out cases served identical customer sets, so its cost result describes a narrow subset.

EXPANDED

Wider route and beam limits and ranked moves add search depth. In the small clustered screen it served all 18 requests and saved 15 dollars per matched stream, compared with 12.50 dollars for `BOUNDED`. Its screen mean case p95 rose from `BOUNDED`'s 3.4 seconds to 20.0 seconds. No held-out, stress, or browser test was retained, so a larger search budget has no demonstrated dispersed-service benefit here.

RUIN_RECREATE

This adds complete reconstruction after removing related visits from `EXPANDED`'s search. It saved 20 dollars per matched clustered stream, a further five dollars over `EXPANDED`, and served all 18 screen requests. Mean case p95 was 23.4 seconds. The reconstruction hypothesis is promising for cost but lacks dispersed and concurrent coverage; its screen result does not establish the same gain under field-like routing.

SHARED

Shared route evaluation reduced screen mean case p95 to 19.5 seconds while matching `RUIN_RECREATE`'s 20 dollar clustered savings. In held-out conditions, equal-customer pairs saved 31.36 dollars per stream, but total service fell to 334 of 480 and 153 searches were incomplete. Mean case p95 reached 64.1 seconds, and the original audit returned HTTP 409. Cost improvement on the matched subset is insufficient evidence for promotion while service and audit behavior remain unresolved.

Daily solver comparison

CURRENT_CAPPED

Relocation and swap stop after the cap, producing about three seconds elapsed at every budget. This is the speed reference when a full solver budget is unavailable. Its paired cost was 1.18, 4.80, and 4.79 dollars worse than `TABU` at 15, 30, and 60 seconds.

CURRENT_UNCAPPED

The same move family without the step cap consumed the full budget and improved paired cost against TABU by 6.38, 3.68, and 3.99 dollars. The capped comparison suggests that additional work can pay on these fixtures, while the two held-out seeds leave its generality open.

LATE_ACCEPTANCE_CHANGE

Relocation-only late acceptance improved as time increased, from 21.85 dollars worse than TABU at 15 seconds to 0.74 dollars worse at 60 seconds. Its trajectory shows that this configuration needs time to approach the baseline; the result does not show an advantage at the tested budgets.

LATE_ACCEPTANCE

Adding swap to the late-acceptance configuration improved its paired result relative to change-only at every budget. It was 13.09 dollars worse than TABU at 15 seconds, near even at 30, and 5.48 dollars better at 60. This conditional comparison is informative because the acceptor is shared; it still has only two independent held-out seeds.

TABU

Relocation and swap with a tabu acceptor are the production reference. Accepted cleanup averaged 55.64 dollars at 15 seconds and 59.26 dollars at 30 and 60, with zero reported violations. Its flat 30-to-60-second mean suggests limited additional benefit on these fixtures, while other configurations may need the longer time to catch up.

SUBLIST

Adding sublist movement and reversal to late acceptance yielded a 6.67 dollar paired advantage at 60 seconds after trailing TABU by 15.06 and 6.68 dollars at 15 and 30. This favors a longer-budget follow-up, not a claim that one specific move caused the gain.

KOPT

The sublist configuration plus k-opt had the largest 60-second paired advantage in this matrix, 8.38 dollars, after deficits of 15.36 and 3.35 dollars at shorter budgets. It is a candidate for replicated long-budget tests because its advantage appears only after consuming the full 60 seconds.

RUIN_RECREATE

Adding list ruin/recreate to the k-opt configuration improved its own accepted cleanup from 15.66 to 34.85 dollars as the budget rose, but it trailed TABU by 39.98, 36.72, and 24.41 dollars. The bigger neighborhood did not repay its search cost at these budgets and fixture sizes.

Report design decisions

Modeled cost leads because the decision concerns operating efficiency. Served requests, feasibility, and elapsed time appear beside it because a cheaper plan that serves fewer customers or misses a search deadline is not an interchangeable outcome. Booking final inventory totals include existing ten-day work and differing served requests, so cost savings use only identical served customer indices within the same seeded dataset. Changed-customer counts remain visible as a guard against selecting only favorable pairs.

Stage labels remain separate because original cases use a bundled earlier policy, screen geography repeats, held-out booking roads are dispersed, stress adds near-capacity concurrent behavior, and browser tests cover only INSERTION. No plotted line fills an untested cell. Daily paired savings use TABU as the retained default and average fleet sizes within each of the two held-out seeds before calculating intervals. The decision matrix does not combine these measures into an unsupported score; it identifies the evidence needed for the next test.

Pros and cons: booking search

Variant	Pros	Cons / limitation
INSERTION	Fast control; held-out 346/480 served with 2.9 s mean case p95; browser 141/240 served.	Higher matched-customer modeled cost than bounded or shared; stress served 17/40.
BOUNDED	Held-out 360/480 served and matched-pair savings of \$23.44 per stream; 21/40 stress served.	Held-out mean case p95 15.7 s; 120 incomplete searches; changed customer sets in 33/48 comparisons.
EXPANDED	Screen matched-pair savings \$15 per clustered stream; 18/18 served.	Screen only; 20.0 s mean case p95; no held-out or stress evidence.
RUIN_RECREATE	Screen matched-pair savings \$20 per clustered stream; 18/18 served.	Screen only; 23.4 s mean case p95; reconstruction cost and no independent held-out evidence.
SHARED	Screen tied ruin/recreate at \$20 matched-pair savings with lower screen latency; held matched pairs show \$31.36 per stream.	Held-out service fell to 334/480, latency reached 64.1 s mean case p95, and original SHARED audit failed.

Pros and cons: daily solver

Configuration	Pros	Cons / limitation
CURRENT_CAPPED	About 3 s runtime, zero violations.	Slightly worse paired cost than TABU at all held budgets.
CURRENT_UNCAPPED	Paired cost better than TABU by \$6.38, \$3.68, \$3.99 at 15/30/60 s.	Consumes full budget; only two held-out seeds.
LATE_ACCEPTANCE_CHANGE	Simple relocation-only comparison, zero violations.	Worse paired cost at 15 and 30 s; near TABU at 60 s.
LATE_ACCEPTANCE	Improves with time; \$5.48 paired advantage at 60 s.	Worse at 15 s and no clear 30 s advantage.
TABU	Production baseline; \$55.64 to \$59.26 mean accepted cleanup; zero violations.	Consumes full budget; alternatives sometimes have lower modeled cost.
SUBLIST	\$6.67 paired advantage at 60 s.	Worse at 15 and 30 s; combined move effect is not isolated.
KOPT	\$8.38 paired advantage at 60 s, largest in this held matrix.	Worse at 15 and 30 s; consumes full 60 s.
RUIN_RECREATE	Improves accepted cost as budget rises; zero violations.	Worse paired cost than TABU at all budgets, by \$24.41 at 60 s.

Decision matrix for next tests

Candidate	Cost signal	Service / feasibility	Speed	Confidence / next test
Booking BOUNDED	Positive matched cost	Best held served, 360/480	15.7 s case p95	Repeat dispersed concurrent streams and inspect incomplete results
Booking SHARED	Positive matched cost	334/480 held served; failed audit	64.1 s case p95	Investigate audit conflict before any promotion
Daily CURRENT_UNCAPPED	Positive at 15/30/60 s	0 violations	Full budget	Repeat on independent geographies and time budgets
Daily KOPT	Best 60 s paired signal	0 violations	60 s	Confirm on more seeds and operational workloads
Daily TABU	Reference	0 violations	Full budget	Retain baseline until replicated evidence

The matrix presents tradeoffs, not a winner. Cost is measured on modeled fixtures; booking service and latency can outweigh a matched-pair cost signal. Feasibility checks passed for validated cases, while a separate audit failure remains unresolved.

Evidence and limits

Source files: summary.json, experiment-manifest.json, validation.json, and field validation report. The generator verifies every inventory SHA-256, raw case counts, the retained failure, and all exclusions. All 218 booking cases are independently validated and all 240 daily cases report zero violations.

The original SHARED seed-83, 50-technician, concurrency-five cold run returned HTTP 409 at final audit. Its client timing and detailed reason were not retained. A separately labeled recovery run passed, and the original failure remains in the archive. Eight original stress-normal-only audits omitted overflow dates and are excluded; reruns with full served dates form the validated stress set. Neither issue is silently counted as a pass.

Browser testing used insertion alone; API latency and polled progress do not measure production browser performance under concurrency. Clustered screen geometry repeats and has zero road seconds. The two held-out seeds limit generalization. The four-hour policy, one-offer selection, and durable lifecycle change together between original and screen; that comparison cannot isolate an algorithm effect. No distance total or production savings is inferred.